6 Related Work

Full-Sequence Output Moderation. Most deployed guardrails operate *post hoc*: they inspect a complete prompt or response after it has already been produced. Models such as LlamaGuard (Inan et al., 2023) and WildGuard (Han et al., 2024) cast safety as full-sequence classification over a fixed taxonomy of harms. These systems are effective as offline judges, but they do not natively solve the streaming problem because they require the completed response before producing a decision. Recent moderators such as BingoGuard (Yin et al., 2025) and policy-adaptive systems such as DynaGuard (Hoover et al., 2025) and OpenAI's gpt-oss-safeguard (OpenAI, 2025) increase granularity or flexibility, but they still rely on full context and therefore inherit the same latency bottleneck in streaming settings.

White-Box and Internal Defenses. A separate line of work reduces moderation latency by leveraging the generator LLM's internal activations. ShieldHead (Xuan et al., 2025) and related internal probes attach lightweight safety heads to hidden states, while recent streaming white-box systems such as Kelp (Li et al., 2025) extend this idea to token-level intervention during decoding by reading internal activations online and modeling risk over time. These methods can be fast, but they are architecture-dependent: the detector is trained on the representation space of a particular base model and is therefore coupled to that model family and checkpoint. In contrast, StreamGuard is a text-only guardrail trained and deployed over prefixes in text space, which makes it compatible with standard black-box moderation setups.

Streaming Safety Systems. Streaming-specific safety systems are the closest prior work to our setting because they moderate partial output prefixes online. Constitutional Classifiers (Sharma et al., 2025) are conceptually close, as they also train on partial outputs to anticipate eventual harmfulness. However, they use prefix-to-final-label supervision, whereas StreamGuard supervises each prefix with a Monte Carlo estimate of the expected harmfulness of likely future continuations. Because the system is not publicly released, we treat it as related work rather than as a direct empirical baseline. In the open text-based setting, Qwen3Guard-Stream (Zhao et al., 2025) is the strongest directly comparable baseline, with publicly released models; however, it uses a different supervision signal, casting streaming moderation as boundary detection by identifying the earliest token at which the response becomes unsafe, labeling earlier prefixes as safe and the boundary and all later prefixes as unsafe or controversial. StreamGuard differs from both approaches in its training target: rather than assigning each prefix the label of one realized completion or the label induced by an explicit unsafe boundary, it supervises each observed prefix with a Monte Carlo estimate of the expected harmfulness of likely future continuations. As a result, StreamGuard is trained to forecast future risk from the current prefix, not merely to inherit a final-output label or detect that an unsafe boundary has already been crossed.

7 Conclusion

We introduced StreamGuard, a unified text-based guardrail for input moderation and streaming output moderation. For streaming outputs, StreamGuard treats moderation as a forecasting problem: given a partial prefix, it predicts the harmfulness of likely future continuations rather than waiting until an explicit violation has already appeared. This enables earlier intervention during generation without requiring exact boundary annotations.

Across standard moderation benchmarks, StreamGuard remains competitive as a conventional moderator while improving over strong published streaming baselines on output-side and intervention-timing evaluations. On the streaming benchmark, it achieves higher overall recall, more on-time interventions, and fewer missed unsafe responses than the comparable boundary-based baseline. We also find that forecasting-based supervision transfers well across tokenizer families, yielding strong output-moderation and streaming performance on Gemma and Qwen backbones. Overall, our results suggest that forecasting is a useful and practical framing for streaming safety: it is simple to train, compatible with text-only black-box moderation, and also effective in low-latency deployment settings.

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